VALVULAR HEART DISEASE

Overview

- Valve either **stenotic** (obstructs forward flow) or **regurgitant** (backward leak).
- **Developing countries** → **rheumatic heart disease (RHD)** commonest; **industrialized** → **degenerative**. Prevalence ↑ with age (13.3% in ≥75).
- **SOB** = common symptom; acute severe SOB = valve failure.

Acute Rheumatic Fever (ARF)

- Children/young adults **5–15 yrs**. Endemic in South Asia, Africa, South America.
- **Pathogenesis**: immune-mediated delayed response to **group A streptococcus** — antigens cross-react with cardiac myosin/sarcolemma. **Aschoff nodules = pathognomonic** (heart only, subacute/chronic phase).
- **Clinical (2–3 wks post-strep pharyngitis)**: fever, arthritis (75%), carditis, rashes, nodules, chorea.
- **Jones criteria**: 2 major OR 1 major + 2 minor + evidence of preceding strep.
- **Carditis**: pancarditis; **soft systolic murmur (MR) very common**; **Carey Coombs murmur** (soft mid-diastolic, valvulitis); AR in 50%; tricuspid/pulmonary rarely.
- **Arthritis**: commonest major; **migratory, asymmetric, large joints** (knees/ankles/elbows/wrists), red/swollen/tender.
- **Skin**: **erythema marginatum** (<5%, trunk/proximal, never face), **subcutaneous nodules** (5–7%, extensor surfaces, >3 wks, confirm not make dx).
- **Sydenham's chorea (St Vitus dance)**: late (≥3mo), females, choreiform movements; ¼ develop chronic valve disease.
- **Investigations**: ESR/CRP (monitor), throat culture (+ve only 10–25%), **ASO titres**, echo (MR, anterior MVL prolapse).

ARF management

- **Bed rest** until temp/WBC/ESR settle.
- **Antibiotics**: single **benzathine benzylpenicillin IM** or oral penicillin ×10d to eradicate strep; penicillin-allergic → erythromycin/cephalosporin.
- **Long-term prophylaxis** (oral phenoxymethylpenicillin BD or monthly benzathine IM): until **age 21** if no carditis; **until 10 yrs after last episode or age 40** (whichever later) if residual heart disease. (Does NOT protect against IE.)
- **Aspirin** for arthritis (response in 24h confirms dx). **Glucocorticoids** for carditis/severe arthritis (faster than aspirin).

Chronic RHD

- Develops in ≥50% of those with carditis; **2/3 women**. Progressive **fibrosis**. **MV affected >90%**, then aortic, tricuspid, pulmonary. **Isolated MS ~25%; mixed MS+MR 40%**.

Mitral Stenosis (MS)

- **Almost always rheumatic.** Normal orifice ~5 cm^2; symptoms when <2 cm^2; severe <1 cm^2.
- **Pathophysiology**: ↑LA pressure → pulmonary venous congestion + SOB. Tachycardia/exercise/pregnancy poorly tolerated. **AF common** (LA dilatation) → can precipitate pulmonary oedema. PHT → RVH/RV dilatation → TR/right HF.
- **Clinical**: effort dyspnoea (dominant), fatigue, haemoptysis, **thromboembolism (esp AF)**.

- **Signs**: **loud S1 (tapping apex)**, **opening snap** (closer to S2 = more severe), **low-pitched mid-diastolic murmur** with **pre-systolic accentuation**, malar flush.
- **Management**: medical (anticoagulation, rate control digoxin/β-blocker for AF, diuretics). **Balloon valvuloplasty = treatment of choice** if criteria met; **MVR** if substantial MR or rigid/calcified valve. (DOACs not approved; IE prophylaxis no longer routine.)

Mitral Regurgitation (MR)

- **Chronic**: gradual LA dilatation, few symptoms, slow LV dilatation (volume overload). **Acute**: rapid LA pressure rise → marked deterioration.
- **MVP (floppy valve)**: common cause of mild MR; myxomatous degeneration; ± Marfan. **Mid-systolic click** ± late systolic murmur; chordal rupture → sudden severe MR; benign arrhythmias, good prognosis.
- **Management**: medical (diuretics, vasodilators, ACE/ARB; digoxin/anticoagulant if AF). **MV repair = treatment of choice for severe MR** (prevents irreversible LV damage); annuloplasty ring if CABG.

Aortic Stenosis (AS)

- Causes by age: **congenital (infancy), bicuspid (young-middle age, 50–60), rheumatic (30–60, + MV disease), calcific/degenerative (70–90)**.
- **Pathogenesis**: ↑pressure gradient → **LVH** → angina (even without CAD) → fixed CO → eventual LV failure.
- **3 cardinal symptoms**: **angina, SOB, syncope** (exertional). Asymptomatic for years then rapid deterioration; **untreated die within 3–5 yrs** of symptoms.
- **Signs**: **harsh ejection systolic murmur radiating to neck**, **soft S2**, S4; **slow-rising low-amplitude pulse (pulsus parvus et tardus)**, narrow pulse pressure, precordial thrill; Gallavardin (radiates to apex).
- **Investigations**: **echo (pivotal) —** gradient/severity; low-flow AS if poor LV; ECG LVH + strain; CT for calcification.
- **Management**: asymptomatic → monitor (every 1–2 yrs, 3–6mo if heavily calcified), treat risk factors. **Symptomatic severe AS** → **prompt AVR** (delay → sudden death). **TAVI** (transfemoral, crushes native valve) for high/inoperable risk → now extreme-to-low risk. Old age NOT a contraindication. Balloon valvuloplasty only useful in congenital.

Aortic Regurgitation (AR)

- Causes: cusp disease, infection, trauma, **aortic root dilatation**. Developing → RHD; developed → root dilation, calcific, bicuspid.
- **Pathogenesis**: LV dilates (volume overload), SV may double → pulsatile arteries → eventually LVF.
- **Clinical**: palpitation, SOB, PND, angina. **Early diastolic murmur** (left sternal edge, held expiration); systolic flow murmur (↑SV, not AS); **Austin Flint murmur** (functional MS). **Acute severe AR** (endocarditis cusp tear) → HF predominates, signs masked.
- **Investigations**: **Doppler echo (1st choice)**; catheterisation/aortography; MRI for aortic dilation.
- **Management**: treat cause (IE/syphilis). **AVR if symptomatic** or **EDD ≥55 mm**; annual echo follow-up; vasodilator (nifedipine) for HTN; aortic root replacement if Marfan/root dilatation.

Tricuspid Stenosis (TS)

- Most commonly **rheumatic** (with MS/AV disease). Systemic venous congestion: **ascites, peripheral oedema, hepatomegaly, prominent a wave, Kussmaul sign**. Low-frequency diastolic murmur at left lower sternal border. **TTE; severe = valve area ≤1.0 cm²**. Treat diuretics, balloon valvotomy or surgery.

Tricuspid Regurgitation (TR)

- Usually **functional** (RV dilatation from right/biventricular HF). **'Giant' v wave in JVP**, pansystolic murmur left sternal border, **pulsatile liver**, oedema, hepatomegaly. Often improves when HF treated; annuloplasty ring if undergoing MVR.

Pulmonary Stenosis (PS)

- Almost always **congenital** (± TOF, Noonan, rubella) or carcinoid. Mild = asymptomatic; moderate → RV overload, exertional dyspnoea; severe → RV failure, cyanosis.
- **Ejection systolic murmur left upper sternum → left shoulder**, ejection click, wide split S2; severe → inaudible P2, **fixed split S2**, RV heave.
- **Doppler echo definitive**. Severe (gradient >50 mmHg) → **balloon valvuloplasty** (excellent results).

Pulmonary Regurgitation (PR)

- Rare isolated; usually PA dilatation from PHT. **Graham Steell murmur** (early diastolic, hard to distinguish from AR, complicates MS). Trivial PR = normal finding.

Prosthetic valves

- **Mechanical** (ball & cage, tilting disc, bileaflet): durable but **require lifelong anticoagulation** (thromboembolism/thrombosis risk); produce clicks.
- **Biological** (pig/allograft): **no anticoagulation needed** but less durable (degenerate 7–10 yrs, worse in mitral); **best for >65 / aortic position**.
- **TAVI**: bioprosthetic via femoral, native valve compressed; avoids sternotomy, short recovery, for high-risk/elderly. **Complications: stroke 2%, heart block needing pacemaker 5–15%.**
- **Prosthetic dysfunction**: unexplained HF → urgent assessment. Mechanical → strut fracture/thrombosis. Biological → regurgitant murmur 8–10 yrs.

High-yield one-liners

- **ARF: Jones criteria, group A strep, Aschoff nodules, migratory arthritis, Sydenham's chorea; penicillin + prophylaxis.**
- **MS = rheumatic; loud S1 + opening snap + mid-diastolic murmur + AF; balloon valvuloplasty.**
- **MR severe → MV repair (treatment of choice). MVP = mid-systolic click.**
- **AS: angina/SOB/syncope; ejection systolic → neck, soft S2, slow-rising pulse; symptomatic severe → AVR/TAVI; die 3–5 yrs untreated.**
- **AR: early diastolic murmur, Austin Flint, collapsing pulse; AVR if symptomatic or EDD ≥55mm.**
- **TR = giant v wave + pulsatile liver. PS = congenital, balloon valvuloplasty.**
- **Mechanical valve = anticoagulate forever; bioprosthetic = no anticoagulation but less durable (>65 yrs).**

VALVULAR HEART DISEASE — Clinical Revision Table

Bold = high-yield. RHD = commonest in developing world; degenerative in industrialized.

RHEUMATIC FEVER

Item	Key points
Cause	Immune response to **group A strep**; cross-reacts cardiac myosin. **Aschoff nodules pathognomonic**
Jones criteria	2 major OR 1 major + 2 minor + preceding strep. Major: **carditis, arthritis, chorea, erythema marginatum, SC nodules**
Carditis	Pancarditis; soft systolic (MR) common; **Carey Coombs** (mid-diastolic valvulitis); AR 50%
Arthritis	**Migratory, asymmetric, large joints**; commonest major
Chorea	Sydenham's (late ≥3mo, females); ¼ → chronic valve disease
Skin	**Erythema marginatum** (trunk, never face); **SC nodules** (extensor, confirm dx)
Treatment	Bed rest; **benzathine penicillin** + long-term prophylaxis; **aspirin** (arthritis), **steroids** (carditis)

LEFT-SIDED STENOSIS & REGURGITATION

Lesion	Cause	Murmur / signs	Symptoms	Management
Mitral stenosis	**Almost always rheumatic**	**Loud S1 (tapping apex), opening snap, mid-diastolic murmur + pre-systolic accentuation**, malar flush	Dyspnoea, fatigue, **AF**, haemoptysis, emboli	Anticoagulate + rate control; **balloon valvuloplasty** (TOC); MVR if calcified
Mitral regurgitation	RHD, MVP, degenerative, ischaemic	**Pansystolic murmur** → **axilla**, soft S1. **MVP = mid-systolic click**	Often few (chronic); acute = sudden deterioration	Diuretics/vasodilator; **MV repair = TOC for severe**
Aortic stenosis	Bicuspid (young), calcific (old), rheumatic	**Harsh ejection systolic** → **neck, soft S2, S4; slow-rising pulse (parvus et tardus)**, thrill	**Angina, SOB, syncope** (exertional); die 3–5yr untreated	Symptomatic severe → **AVR / TAVI**; balloon only congenital
Aortic regurgitation	RHD, root dilation, bicuspid, IE	**Early diastolic murmur** (LSE, expiration); **Austin Flint**; collapsing pulse, wide pulse pressure	Palpitation, SOB, PND, angina	**AVR if symptomatic or EDD ≥55mm**; root replacement if Marfan

RIGHT-SIDED & PULMONARY

Lesion	Key points
Tricuspid stenosis	Rheumatic (+ MS); systemic congestion (**ascites, hepatomegaly, oedema, prominent a wave, Kussmaul**); severe ≤1.0 cm²; diuretics/valvotomy
Tricuspid regurgitation	Usually **functional** (RV dilatation/right HF); **giant v wave, pulsatile liver**, pansystolic LSE; improves with HF treatment; annuloplasty if MVR
Pulmonary stenosis	Almost always **congenital** (± TOF, Noonan); ejection systolic + click → left shoulder; severe = **fixed split S2**; **balloon valvuloplasty** if gradient >50
Pulmonary regurgitation	Rare; PHT/PA dilatation; **Graham Steell murmur** (mimics AR); trivial PR normal

PROSTHETIC VALVES

Type	Key points
Mechanical	Durable; **lifelong anticoagulation** (thromboembolism/thrombosis); clicks; strut fracture
Biological	**No anticoagulation needed** but less durable (degenerate 7–10yr, worse mitral); **best >65 / aortic**
TAVI	Bioprosthetic via femoral; native valve compressed; no sternotomy; high-risk/elderly. **Stroke 2%, heart block needing pacemaker 5–15%**